

Assignment - 4

CSL 2050 - Pattern Recognition & Machine Learning

Max. Marks: 50

Submission Instructions:

- Students must submit one Jupyter notebook (.ipynb) file containing:
 - Executable code
 - Visible outputs (tables, figures, metrics)
 - Brief comments or markdown explanations where required
- File naming format : **RollNo.-Name.ipynb** example, 21CS045-AmitSharma.ipynb
- Do Not submit PDFs, word files, or screenshots.
- Late submissions will be penalized as per course policy.
- Date of last submission: **15th April 2026, 11:59PM.**

Part A: Decision Trees [use library]

Dataset: Breast Cancer Wisconsin

<https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data/code>

Tasks:

1. Perform exploratory analysis i.e., evaluating data distribution, printing feature shape, class distribution. [5]
2. Implement Impurity measures i.e., Gini index and entropy [5]
3. Information Gain: [5]
 - A. For each feature, try possible thresholds and compute impurity reduction
 - B. Return best feature and threshold

4. Implement the decision tree (using library) in order to create a generalized classification algorithm (that does not overfit). Visualize the decision tree and its boundary [5]
5. Prediction and Evaluation [5]
 - A. Predict labels for test data
 - B. Compute accuracy. Visualize the results via confusion matrix

Part B: Support Vector Machines [use library]

Dataset: Same as above (Breast Cancer Wisconsin)

Tasks:

1. Data Preparation Pipeline: [5]
 - A. Split dataset into train/test sets
 - B. Perform feature scaling using StandardScaler
 - C. Verify shapes of processed data
2. Train an SVM with linear kernel [use library]. Showcase the number of support vectors, model coefficients (if available). [5]
3. Model Evaluation: [5]
 - A. Predict on test set
 - B. Compute accuracy and confusion matrix
4. Kernel Comparison: [5]
 - A. Train SVM with RBF kernel.
 - B. Compare the accuracy of the RBF kernel with the linear kernel.
5. Hyperparameter Tuning: [5]
 - A. Experiment with parameters i.e., C (regularization) and gamma (for RBF)
 - B. Create a table showing parameter values vs accuracy.